Data Sets

There are many different datasets specifically for TTS in deep learning. Most (if not all) I could find utilized a supervised model, matching the spoken audio to the synched text transcript. One dataset I came across was LJSpeech. It has around 13,100 data pairs (sound files and transcripts), and uses one speaker for all the files, with around 24 hours of audio. This would be beneficial for creating a consistent voice. However, it also seems to use a lot of higher level vocabulary, which for my project wouldn't be necessary, since I would want to focus on communication for middle to elementary aged kids.

Multilingual LibriSpeech was another dataset I found that also offered soundbytes in other languages, acting as an extension of the LibriSpeech dataset which is in English which has about 1000 hours of data. I could utilize them in conjecture to potentially add a multilingual compatibility. This is something that I was thinking of since the overall scope of my original project seems like it wouldn't be too difficult considering the large amounts of research that already exist on the topic. I could also start with just the English files, and if I have the capability to add other languages, do that after the fact. The biggest stipulation is that LibriSpeech was originally created with the intention of turning audio into text. The dataset could be used to do the reverse, but may require some extra steps.

The last dataset I looked at was Libri-TTS/ LibriTTS-R. It uses multiple different speakers, varying in accent. The audio has a total runtime of about 585 hours. It also has the capability to view the audio in context, meaning I could use it to mimic that cadence in the final product. This also means that I would have more diverse data to work with, but that it would be more challenging to get the best end result.

Given all of this information, I'm thinking either LibriSpeech or Libri-TTS/ LibriTTS-R would be the best option. While LibriSpeech has more hours of data, as well as the multilingual addon, the audio quality isn't as high as Libri-TTS/ LibriTTS-R. LibriSpeech, as I mentioned, also wasn't built for turning text into audio, which is the focus of my project. Libri-TTS/ LibriTTS-R also already has the preprocessing done for you. So, it is for these reasons that the dataset I will focus on will be **Libri-TTS/ LibriTTS-R**.